

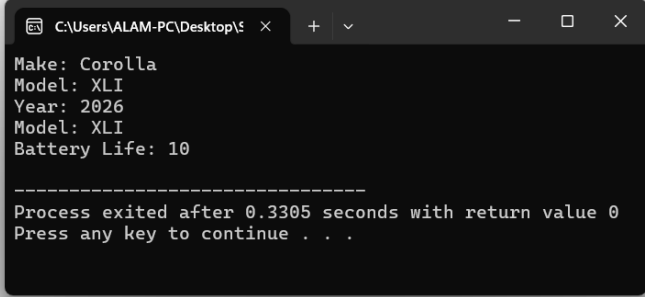
OOP LAB 4

SOLVED

Walkthrough Task:
There is no walkthrough.

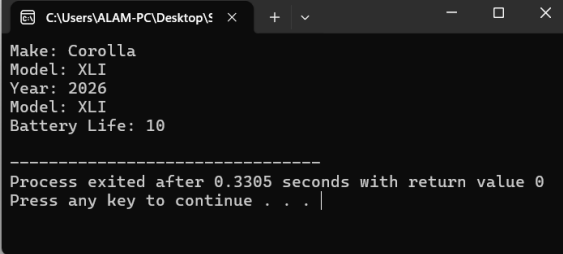
Practice Task 1:

```
PT1.cpp
1  #include<iostream>
2  using namespace std;
3  class Car{
4  private:
5      int year;
6  protected:
7      string model;
8  public:
9      string make;
10     void setmake(string m){
11         make = m;
12     }
13     void setmodel(string mo){
14         model = mo;
15     }
16     void setyear(int y){
17         year = y;
18     }
19     string getmake(){
20         return make;
21     }
22     string getmodel(){
23         return model;
24     }
25     int getyear(){
26         return year;
27     }
28     void displayvalues(){
29         cout<<"Make: "<<make<<endl;
30         cout<<"Model: "<<model<<endl;
31         cout<<"Year: "<<year<<endl;
32     }
33 };
34 class ElectricCar : public Car{ // Derived class must be written OUTSIDE the base class
35 private:
36     int batterylife;
37 public: // Always use public: for member functions so they can be called in main()
38     void setbatterylife(int b){
39         batterylife = b;
40     }
41     int getbatterylife(){
42         return batterylife;
43     }
44     void showmodel(){
45         cout<<"Model: "<<model<<endl;
46     }
47 };
48 int main(){
49     ElectricCar c1; // Create object of derived class ? it can use BOTH Car and ElectricCar functions
50     c1.setmake("Corolla"); // if you write ; ; 2 semicolon it is not error ( ; means an empty statement )
51     c1.setmodel("XLI");
52     c1.setyear(2026);
53     c1.displayvalues();
54     c1.setbatterylife(10);
55     c1.showmodel();
56     cout<<"Battery Life: "<<c1.getbatterylife()<<endl;
57     return 0;
58 }
59
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66
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68
69
```



```
Make: Corolla
Model: XLI
Year: 2026
Model: XLI
Battery Life: 10

-----
Process exited after 0.3305 seconds with return value 0
Press any key to continue . . .
```



```
Make: Corolla
Model: XLI
Year: 2026
Model: XLI
Battery Life: 10

-----
Process exited after 0.3305 seconds with return value 0
Press any key to continue . . .
```

Practice Task 2:

```
PT1.cpp PT2.cpp
1 #include<iostream>
2 using namespace std;
3 class Animal{ // this is base class
4     private:
5         int age;;
6     protected:
7         string name;
8     public:
9         void setvalues(int a, string n){
10             age = a;
11             name = n;
12         }
13         void display(){
14             cout<<"Dog name: "<<name<<endl;
15             cout<<"Dog Age: "<<age<<endl;
16         }
17 };
18 class Dog : public Animal{ // this is derived class
19     public:
20         void displaydog(){
21             cout<<"Dog name: "<<name<<endl;
22         }
23 };
24 int main(){
25     Dog d1;
26     d1.setvalues(4,"Rocky");
27     d1.display();
28     d1.displaydog();
29     return 0;
30 }
```

```
C:\Users\ALAM-PC\Desktop\5 x + - □ X
Dog name: Rocky
Dog Age: 4
Dog name: Rocky

-----
Process exited after 0.284 seconds with return value 0
Press any key to continue . . . |
```

Practice Task 3:

```
PT1.cpp PT2.cpp PT3.cpp
1 #include<iostream>
2 using namespace std;
3 class Book{
4     private:
5         string title;
6         string author;
7         float price;
8     public:
9         Book(string t, string a, float p){
10             title = t;
11             author = a;
12             price = p;
13             cout<<"Constructor Called"<<endl;
14         }
15         void setprice(float p){
16             price = p;
17         }
18         void getdetail(){
19             cout<<"Title: "<<title<<endl;
20             cout<<"Author: "<<author<<endl;
21             cout<<"Price: "<<price<<endl;
22         }
23         ~Book(){
24             cout<<"Destructor Called for Object "<<endl;
25         }
26 };
27
28 int main(){
29     Book b1("Robinson Crusoe", "Daniel Defoe", 1500);
30     b1.setprice(600); // First constructor sets value, then setPrice updates it.
31     b1.getdetail();
32     return 0;
33 }
```

```
C:\Users\ALAM-PC\Desktop\5 x + - □ X
Constructor Called
Title: Robinson Crusoe
Author: Daniel Defoe
Price: 600
Destructor Called for Object

-----
Process exited after 0.3196 seconds with return value 0
Press any key to continue . . . |
```